
Predictable Token Divergence, Surprising Residuals: Comparing Base and Instruct Models

Ari Holtzman and NeuriCo

Abstract

Instruction tuning changes language models, but the change is not uniform across tokens. We study whether per-token distributional divergence between a base model and its instruction-tuned counterpart is predictable, and whether prediction failures form a useful audit signal. Using real logits from QWEN2.5-1.5B and QWEN2.5-1.5B-INSTRUCT, we compute $\text{KL}(p_{\text{inst}} \parallel p_{\text{base}})$ over 24,049 teacher-forced token positions from ALPACA, JUST-EVAL-INSTRUCT, BEAVERTAILS, CHATBOT ARENA, and WIKITEXT-2. We train gradient-boosted regressors to predict $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$ from source, segment, position, token-surface features, and base-model uncertainty. The rich predictor improves held-out MAE from 0.0601 for a position/source baseline to 0.0460, with a grouped bootstrap 95% confidence interval of $[-0.0163, -0.0118]$ for the rich-minus-position MAE difference. Its residuals are more informative than its aggregate error: high-residual tokens are enriched for generation starts ($5.17\times$), assistant markers ($4.78\times$), refusal-style tokens ($4.54\times$), code/math tokens ($3.49\times$), and WIKITEXT-2 tokens ($3.13\times$). The largest positive residuals unexpectedly concentrate in one natural-text narrative span, while large negative residuals often occur at explicit Assistant boundaries where both models agree. These results support residual analysis as a compact triage tool for model-difference auditing.

1 Introduction

Instruction tuning makes a base language model easier to use, but it does not change every next-token distribution in the same way. Some changes are expected: an instruct model should be more ready to start an answer, follow chat markup, and refuse unsafe requests. Other changes are harder to anticipate. The practical question is not only where two models differ, but where they differ more or less than ordinary surface cues predict.

This question matters for alignment auditing and model diffing. A researcher comparing a base model with its instruct version can compute token-level divergences, but a long table of logits is not an explanation. If ordinary factors such as source, position, prompt boundary, token class, and base uncertainty explain much of the divergence, then the remaining residuals are a small, inspectable set of cases where the post-training effect is surprising.

Prior work shows that many base-to-chat changes are concentrated around stylistic tokens, response openings, and safety phrases [Lin et al., 2023, Ghosh et al., 2024, Qi et al., 2024]. However, most analyses report aggregate shifts or pre-defined token categories. Our main question is different: **after learning the obvious regularities, what remains surprising?**

We propose RESIDUAL-KL, a simple residual-auditing workflow. We score a same-family base/instruct pair on identical teacher-forced contexts, train predictors of $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$, and inspect held-out residuals. The workflow is summarized in figure 1. It treats prediction as a filter rather than an end goal: accurate predictions tell us which shifts are ordinary, and large residuals tell us where to look next.

Our experiment uses QWEN2.5-1.5B and QWEN2.5-1.5B-INSTRUCT with a shared 151,936-token vocabulary. Across 299 encoded examples and 24,049 token positions, the rich predictor achieves 0.0460 held-out MAE, compared with 0.0601 for a position/source baseline and 0.0756 for a mean baseline. The improvement over the position/source baseline is stable under 500 grouped bootstrap resamples. More importantly, high residuals recover known high-shift regions and surface an unexpected WIKITEXT-2 narrative cluster: 84 of the top 150 absolute residuals are WIKITEXT-2 tokens, and 61 come from one example.

Our contributions are:

- We measure full-vocabulary, teacher-forced token divergence between a real base model and its instruction-tuned counterpart across instruction, safety, chat, and natural-text sources.
- We train and evaluate KL-divergence predictors under an example-grouped split, showing that a rich feature set improves MAE by 0.0142 over a position/source baseline.
- We analyze residual failures quantitatively and qualitatively, showing that prediction misses are enriched for interpretable token classes and for an unexpected natural-text outlier.
- We identify a key audit lesson: top-1 agreement can hide large full-distribution shifts, so residual analysis should inspect distributional divergence rather than only argmax changes.

The rest of the paper reviews related work in instruction tuning and model diffing (section 2), defines the scoring and prediction procedure (section 3), reports quantitative and residual results (section 4), and discusses implications and limitations (section 5).

2 Related Work

Instruction tuning and superficial shifts. Instruction tuning and RLHF are standard ways to turn pretrained language models into assistants [Wei et al., 2022, Ouyang et al., 2022, Chung et al., 2022, Longpre et al., 2023]. These methods improve usability, but recent analyses show that many changes can be sparse and stylistic. Lin et al. [2023] compare base and aligned models at the token-distribution level and argue that alignment often leaves top choices unchanged while shifting discourse and safety style. Ghosh et al. [2024] similarly find that instruction tuning can emphasize response-initiation and style tokens, and that full-parameter tuning can borrow surface patterns from instruction data. We build on this line of work by treating those expected shifts as predictable structure, then studying residual failures.

Prompt-based alignment controls. Prompting and in-context examples can imitate part of instruction tuning. URIAL aligns base models through a system prompt and a few stylistic examples [Lin et al., 2023], and Zhao et al. [2024] show that in-context alignment can approach but not fully replace instruction fine-tuning in many settings. Decoding-time approaches such as proxy tuning and nudging also use tuned-vs-untuned distribution differences as a steering signal [Liu et al., 2024, Fei et al., 2025]. These works motivate future controls for our setting: if a residual cluster appears under both an instruct model and a base-plus-prompt condition, it may reflect prompt format rather than weight-level post-training.

Token-level safety and preference signals. Several alignment methods and analyses operate directly at the token level. Qi et al. [2024] show that safety alignment can be shallow, with divergence concentrated in the first output tokens. Token-level DPO uses per-token KL constraints as part of a preference objective [Zeng et al., 2024]. Refusal-direction work identifies a low-dimensional mechanism mediating refusal behavior in chat models [Arditi et al., 2024]. Our experiment does not claim a mechanistic explanation, but it uses token-level KL and residual enrichment to prioritize spans that such mechanistic tools could inspect.

Model diffing and interpretability. Model diffing studies how fine-tuning changes internal computations. Crosscoders compare base and chat representations, but Minder et al. [2025] show that sparse dictionary methods can misattribute shared concepts as chat-specific without careful controls. This caution applies directly to residual auditing: large KL residuals are discovery targets, not final evidence of a causal mechanism. We therefore report residuals as triage signals and emphasize follow-up with prompt controls, top- k probability inspection, and activation-level validation.

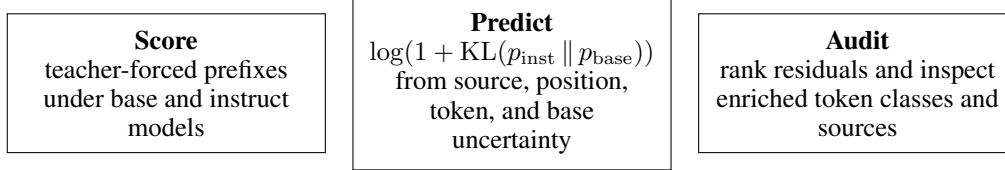


Figure 1: Overview of RESIDUAL-KL. The predictor is not the final object of study. We use it to subtract expected source, boundary, token-class, and uncertainty effects, then inspect the held-out locations where actual KL is much higher or lower than predicted.

Table 1: Datasets used for teacher-forced token scoring. We score 299 encoded examples and 24,049 token positions after tokenization and filtering.

Source	Examples	Token rows	Role in experiment
ALPACA 52k	60	5,119	Instruction-output SFT traces
JUST-EVAL-INSTRUCT	60	1,422	Prompt-only instruction boundary traces
BEAVERTAILS 30k	60	4,972	Safety/refusal-related traces
CHATBOT ARENA mirror	60	7,329	Real chat prompt-response traces
WIKITEXT-2 Raw	59	5,207	Natural-text baseline
Total	299	24,049	Multi-source divergence benchmark

Datasets for divergence analysis. We use datasets that cover distinct contexts where instruction tuning might matter. ALPACA supplies instruction-output traces [Taori et al., 2023]; JUST-EVAL-INSTRUCT comes from the URIAL evaluation setting [Lin et al., 2023]; BEAVERTAILS supplies safety-related prompt-response traces [Ji et al., 2023]; CHATBOT ARENA provides real chat conversations [Zheng et al., 2023]; and WIKITEXT-2 provides natural text outside an instruction setting [Merity et al., 2016]. This mix lets us ask whether residual surprises are restricted to assistant-style data or also appear in ordinary text.

3 Methodology

Problem formulation. For a token context c_t , let $p_{\text{base}}(\cdot | c_t)$ be the base-model next-token distribution and $p_{\text{inst}}(\cdot | c_t)$ be the instruct-model distribution. Our primary target is the forward KL

$$d_t = \text{KL}(p_{\text{inst}} \| p_{\text{base}}) = \sum_{v \in \mathcal{V}} p_{\text{inst}}(v | c_t) \log \frac{p_{\text{inst}}(v | c_t)}{p_{\text{base}}(v | c_t)}. \quad (1)$$

We train predictors on $y_t = \log(1 + d_t)$ because token-level KL is right-skewed. Residuals are

$$r_t = y_t - \hat{y}_t, \quad (2)$$

where positive residuals mean actual divergence is higher than predicted and negative residuals mean the predictor expected more divergence than occurred.

Models. We compare QWEN2.5-1.5B and QWEN2.5-1.5B-INSTRUCT [Qwen Team, 2024]. This pair is useful because the two checkpoints are same-family, open-weight, and share the Qwen tokenizer. We used the Qwen instruct tokenizer, bfloat16 model forward passes, and float32 divergence computation. The final run used `cuda:0`.

Datasets. We score five local datasets, shown in table 1. Traces are capped at 224 tokens. For instruction, safety, and chat sources, we keep prompt and response segments when available. For prompt-only JUST-EVAL-INSTRUCT traces, we also score the generation-start distribution after the prompt.

Token-level scoring. Each encoded trace is evaluated under both models on the same teacher-forced prefix. We save forward KL, reverse KL, Jensen-Shannon divergence, base entropy, base top probability, base top-1 margin, observed-token base log probability and rank, top-1 agreement, and decoded base/instruct top-1 tokens. Full-vocabulary KL is computed from float32 log-softmax values over the shared vocabulary.

Table 2: Source-level forward KL summaries. KL is $\text{KL}(p_{\text{inst}} \parallel p_{\text{base}})$ over teacher-forced next-token distributions. The largest value in each numeric column is bolded.

Source	Mean KL	Median KL	P90 KL	P99 KL	Top-1 disagree
JUST-EVAL-INSTRUCT	0.109	0.063	0.271	0.585	0.187
BEAVERTAILS	0.104	0.043	0.208	1.237	0.150
WIKITEXT-2	0.096	0.042	0.184	1.145	0.131
CHATBOT ARENA	0.084	0.035	0.176	1.260	0.131
ALPACA	0.071	0.031	0.172	0.559	0.128

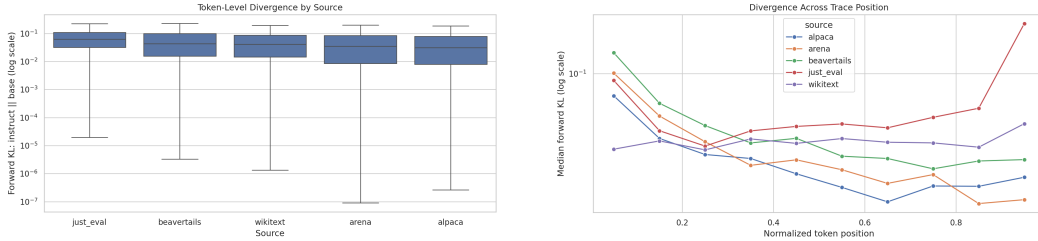


Figure 2: Token-level KL by source and position. (Left) shows that JUST-EVAL-INSTRUCT has the largest mean KL and top-1 disagreement, while ALPACA has the lowest mean KL. (Right) shows that divergence is concentrated near boundaries and early positions rather than being uniform across traces.

Predictors and baselines. All predictors target $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$. The MEAN baseline predicts the training mean. The POSITION/SOURCE baseline uses source, segment, context position, normalized position, response-token index, and a generation-start flag. The RICH PREDICTOR adds token length, whitespace/newline/digit/capitalization flags, observed-token availability, base entropy, base top probability, base margin, observed-token log probability, observed-token rank, token class, frequent token identity, safety label, and frequent safety category. Both learned predictors use HistGradientBoostingRegressor from scikit-learn [Pedregosa et al., 2011].

Evaluation and statistics. We split held-out data by example id, not token row, to avoid putting positions from the same trace in both train and test. The final split has 224 train examples and 75 test examples. We report MAE, RMSE, R^2 , and Spearman correlation on held-out tokens. We estimate uncertainty with 500 grouped bootstrap resamples over held-out examples. For source comparisons, we compute per-example mean KL, run Mann-Whitney U tests, correct with Holm adjustment, and report Cliff’s delta as an effect size.

4 Results

4.1 Source-Level Divergence

table 2 reports divergence by source. Mean forward KL ranges from 0.071 for ALPACA to 0.109 for JUST-EVAL-INSTRUCT. The top-1 disagreement rate follows a similar pattern: JUST-EVAL-INSTRUCT has the largest disagreement rate at 0.187, while ALPACA has the lowest at 0.128. These values confirm that instruction and boundary-heavy sources are more divergent on average, but the differences are not limited to prompt datasets. WIKITEXT-2, a natural-text baseline, has mean KL 0.096 and a P99 KL of 1.145.

Generation-start positions are much more divergent than ordinary continuations. As shown in table 3, BEAVERTAILS has generation-start mean KL 0.802 and P90 KL 2.085, compared with a source-wide mean KL of 0.104. CHATBOT ARENA, ALPACA, and JUST-EVAL-INSTRUCT also show elevated generation-start KL. This supports the view that response boundaries are a major site of instruction-tuning shift.

At the per-example level, the strongest Holm-corrected source differences are ALPACA versus JUST-EVAL-INSTRUCT and ALPACA versus BEAVERTAILS. The ALPACA-vs-JUST-EVAL-INSTRUCT

Table 3: Generation-start divergence by source. Boundary positions are substantially more divergent than average continuations, especially for BEAVERTAILS. The largest value in each metric is bolded.

Source	Mean KL	Median KL	P90 KL
BEAVERTAILS	0.802	0.504	2.085
CHATBOT ARENA	0.444	0.414	0.846
ALPACA	0.355	0.296	0.627
JUST-EVAL-INSTRUCT	0.352	0.272	0.688
WIKITEXT-2	0.159	0.095	0.262

Table 4: Strongest source-level Mann-Whitney comparisons on per-example mean KL after Holm correction. Cliff’s delta is for the first source relative to the second.

Comparison	Mean A	Mean B	Holm p	Cliff’s δ
ALPACA vs. JUST-EVAL-INSTRUCT	0.0898	0.1131	0.000087	-0.471
ALPACA vs. BEAVERTAILS	0.0898	0.1230	0.00710	-0.356

comparison has Holm-corrected $p = 8.66 \times 10^{-5}$ and Cliff’s delta -0.471 ; the ALPACA-vs-BEAVERTAILS comparison has Holm-corrected $p = 0.00710$ and Cliff’s delta -0.356 . Most other pairwise source differences do not remain significant after correction, including CHATBOT ARENA versus WIKITEXT-2.

4.2 Predicting Token-Level KL

table 5 shows that token-level KL is predictable above simple baselines. The POSITION/SOURCE baseline improves MAE from 0.0756 to 0.0601 and explains 27.10% of held-out variance. The RICH PREDICTOR further reduces MAE to 0.0460, improves RMSE to 0.1183, and raises Spearman correlation to 0.795. The grouped bootstrap mean rich-minus-position MAE difference is -0.0142 , with a 95% confidence interval of $[-0.0163, -0.0118]$. Because the interval is entirely below zero, the improvement is stable across held-out-example resampling.

4.3 Residuals Identify Expected and Unexpected Shifts

The rich predictor’s median absolute residual is 0.0202. We define high-residual tokens as the top 2% by absolute residual, corresponding to a threshold of 0.2961. table 6 shows that high residuals recover expected categories: generation starts are enriched $5.17\times$, assistant markers $4.78\times$, refusal-style tokens $4.54\times$, and code/math tokens $3.49\times$. The surprising enrichment is source-level: WIKITEXT-2 tokens are enriched $3.13\times$ among high residuals.

table 7 summarizes representative residual failures. The largest positive residuals are dominated by a single WIKITEXT-2 narrative example. In the top 150 absolute residuals, 84 are WIKITEXT-2 rows, and 61 come from `wikitext:4314`. This is not instruction boilerplate: the snippets are ordinary narrative text, yet the instruct and base distributions sharply diverge. One row has the same decoded top-1 token for both models, but KL is 4.282 and the residual is 1.655, showing that argmax agreement can hide large full-distribution changes.

Large negative residuals often appear near explicit Assistant markers. The predictor learns that assistant markers and generation starts are usually high-divergence zones, but in several held-out contexts both models assign similar distributions to the literal marker. This failure mode is useful: it shows that assistant markup is not uniformly divergent. The local context and whether the boundary is already explicit affect whether the base model agrees with the instruct model.

5 Discussion

Boundary effects are real but partly predictable. Generation starts, prompt/response boundaries, and assistant markers show elevated divergence, especially in BEAVERTAILS. This matches prior evidence that instruction and safety tuning can concentrate changes in early or stylistically important tokens [Lin et al., 2023, Ghosh et al., 2024, Qi et al., 2024]. The predictor results add a second point:

Table 5: Held-out predictor performance on $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$. Lower is better for MAE and RMSE; higher is better for R^2 and Spearman correlation. Best values are bolded.

Model	MAE	RMSE	R^2	Spearman r
MEAN baseline	0.0756	0.1471	-0.0007	n/a
POSITION/SOURCE baseline	0.0601	0.1256	0.2710	0.414
RICH PREDICTOR	0.0460	0.1183	0.3533	0.795

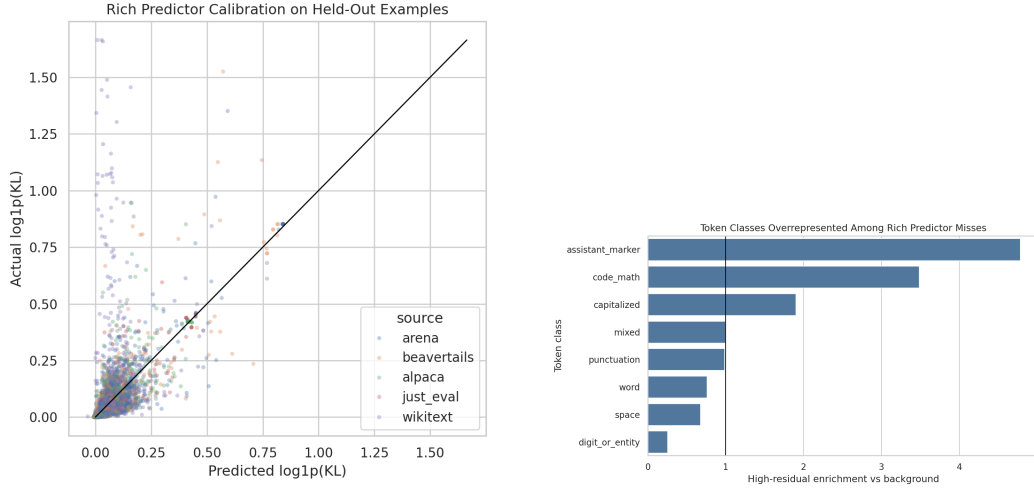


Figure 3: Prediction and residual analysis. (Left) shows held-out rich-predictor predictions against actual $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$. (Right) shows enrichment among the top 2% absolute residuals; known boundary and assistant-style classes are enriched, but WIKITEXT-2 is also strongly overrepresented.

many of these shifts are predictable from source and position alone. This is why residuals are useful. They separate common boundary effects from cases where boundary features over- or under-explain the actual distributional shift.

Natural text can produce surprising residuals. The strongest positive misses were not refusal boilerplate, chat openings, or formatting tokens. They were concentrated in WIKITEXT-2 narrative continuations. This finding does not prove that instruction tuning generally changes narrative modeling, because one example contributes many of the top residuals. It does show that a natural-text source can contain high-KL spans that surface-feature predictors do not expect. This is exactly the type of example residual auditing is designed to find.

Top-1 agreement is too weak for model comparison. Several high-KL residuals have identical base and instruct top-1 tokens. A top-token comparison would mark those positions as agreement, but the full distributions differ sharply. For alignment auditing, this matters because downstream sampling behavior, refusal likelihood, and continuation diversity can depend on probability mass beyond the argmax. Full-distribution metrics such as KL and JS divergence are therefore better suited for triage than top-1 agreement alone.

Limitations. This study uses one model pair at 1.5B parameters. Larger Qwen models, Gemma, Llama, and Mistral pairs may show different residual structure. The dataset sample is intentionally medium-scale: 299 examples and 24,049 token positions are enough to test the workflow, but not enough for a definitive taxonomy. The rich predictor uses observed-token surface features and base-model statistics, which is appropriate for teacher-forced analysis but not a deployment setting where future sampled tokens are unknown. We also do not include a base-plus-URL prompt control, so prompt-format effects and weight-level instruction-tuning effects remain partly entangled. Finally, residual labels are heuristic discovery aids, not mechanistic explanations.

Table 6: Enrichment among high-residual tokens. High residuals are the top 2% of held-out tokens by absolute rich-predictor residual.

Feature	Value	Enrichment
Segment	generation start	5.17 $\times$
Token class	assistant marker	4.78 $\times$
Token class	refusal style	4.54 $\times$
Token class	code/math	3.49 $\times$
Source	WIKITEXT-2	3.13 $\times$
Token class	capitalized	1.90 $\times$

Table 7: Representative rich-predictor residual failures. Residuals are actual minus predicted $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$. Positive residuals indicate underprediction; negative residuals indicate overprediction.

Source	Token	KL	Pred.	Residual	Agree	Base top-1	Inst. top-1	Interpretation
WIKITEXT-2	is	4.282	0.0097	1.655	no	quote	quote	Top-1 text agrees, but probability mass beyond the argmax differs strongly.
WIKITEXT-2	other	4.283	0.0281	1.636	no	two	pool	Natural narrative continuation shift not captured by source/position/token features.
WIKITEXT-2	cares	4.249	0.0334	1.625	no	calls	approaches	The instruct model favors a different narrative continuation mode.
CHATBOT ARENA	Assistant	0.146	0.508	-0.372	yes	Assistant	Assistant	Literal chat markup makes the base model agree more than expected.
BEAVERTAILS	Assistant	0.085	0.405	-0.323	yes	Assistant	Assistant	A usually high-divergence boundary collapses when the marker is explicit.
CHATBOT ARENA	Transform	0.037	0.297	-0.260	yes	Transformers	Transformers	Response start looked risky to the predictor, but both models agree.

Implications. RESIDUAL-KL is best viewed as an audit queue. It does not replace careful qualitative analysis or activation-level validation, but it can focus that work on a small set of surprising positions. The natural next step is to expand natural-text sampling, inspect top- k probability tables for high positive residuals, and test whether the same spans remain surprising under base-plus-prompt controls. After replication, activation-level tools such as refusal-direction probes or crosscoders can test whether high-residual spans correspond to stable internal differences.

6 Conclusion

We asked where a base model and its instruction-tuned counterpart differ token by token, and whether the failures of a learned divergence predictor reveal useful surprises. On QWEN2.5-1.5B versus QWEN2.5-1.5B-INSTRUCT, a rich predictor explains substantially more held-out $\log(1 + \text{KL}(p_{\text{inst}} \parallel p_{\text{base}}))$ than mean and position/source baselines. Its residuals recover known shift zones such as generation starts, assistant markers, refusal-style tokens, and code/math, while also surfacing an unexpected WIKITEXT-2 narrative cluster.

The central takeaway is that predictability and surprise are both useful. Predictable KL tells us which base-vs-instruct changes follow ordinary source, boundary, and token cues. Residual KL tells us where those cues fail, giving researchers a compact set of spans for human inspection and mechanistic follow-up. Future work should replicate the analysis across larger same-family model pairs, add base-plus-prompt controls, and expand natural-text sampling to determine whether the WIKITEXT-2 outlier is isolated or part of a broader post-training shift.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Rimsky, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction, 2024.

- Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, et al. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 2022. URL <https://arxiv.org/abs/2210.11416>.
- Yu Fei, Yasaman Razeghi, and Sameer Singh. Nudging: Inference-time alignment of llms via guided decoding. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2025. doi: 10.18653/v1/2025.acl-long.623.
- Sreyan Ghosh, Chandra Kiran Reddy Evuru, Reddy Evuru, Sonal Kumar, S. Ramaneswaran, Deepali Aneja, Zeyu Jin, Ramani Duraiswami, et al. A closer look at the limitations of instruction tuning. In *Proceedings of the International Conference on Machine Learning*, 2024. doi: 10.48550/arXiv.2402.05119.
- Jiaming Ji, Mickel Liu, Josef Dai, Xuehai Pan, Chi Zhang, Ce Bian, Boyuan Chen, Ruiyang Sun, Yizhou Wang, and Yaodong Yang. BeaverTails: Towards improved safety alignment of llm via a human-preference dataset. In *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2307.04657>.
- Bill Yuchen Lin, Abhilasha Ravichander, Ximing Lu, Nouha Dziri, Melanie Sclar, Khyathi Raghavi Chandu, Chandra Bhagavatula, and Yejin Choi. The unlocking spell on base llms: Rethinking alignment via in-context learning, 2023.
- Alisa Liu, Xiaochuang Han, Yizhong Wang, Yulia Tsvetkov, Yejin Choi, and Noah A. Smith. Tuning language models by proxy. In *Proceedings of the International Conference on Learning Representations*, 2024. doi: 10.48550/arXiv.2401.08565.
- Shayne Longpre, Le Hou, Tu Vu, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc V. Le, et al. The flan collection: Designing data and methods for effective instruction tuning. In *Proceedings of the International Conference on Machine Learning*, 2023. doi: 10.48550/arXiv.2301.13688.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models, 2016.
- Julian Minder, Clement Dumas, C. Juang, Bilal Chughtai, and Neel Nanda. Overcoming sparsity artifacts in crosscoders to interpret chat-tuning, 2025.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022. doi: 10.52202/068431-2011.
- Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. In *Proceedings of the International Conference on Learning Representations*, 2024. doi: 10.48550/arXiv.2406.05946.
- Qwen Team. Qwen2.5-1.5B and Qwen2.5-1.5B-Instruct model cards, 2024. URL <https://huggingface.co/Qwen/Qwen2.5-1.5B>.
- Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model, 2023. URL <https://crfm.stanford.edu/2023/03/13/alpaca.html>.
- Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, et al. Finetuned language models are zero-shot learners. In *Proceedings of the International Conference on Learning Representations*, 2022. URL <https://arxiv.org/abs/2109.01652>.

- Yongcheng Zeng, Guoqing Liu, Weiyu Ma, Ning Yang, Haifeng Zhang, and Jun Wang. Token-level direct preference optimization. In *Proceedings of the International Conference on Machine Learning*, 2024. doi: 10.48550/arXiv.2404.11999.
- Hao Zhao, Maksym Andriushchenko, Francesco Croce, and Nicolas Flammarion. Is in-context learning sufficient for instruction following in llms? In *Proceedings of the International Conference on Learning Representations*, 2024. doi: 10.48550/arXiv.2405.19874.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2306.05685>.